

Dataset Documentation and Data Dictionary

Dataset Overview

The **Campaign_Data_Week1** dataset captures individual customer interactions with NewCo's digital ad campaign across multiple marketing channels and campaign types. Each record represents a single digital interaction and details interaction metrics, customer status, conversion status, and resulting monetary sales value.

- **Format:** Excel Workbook (.xlsx)
- **Primary Key:** Interaction ID
- **Granularity:** One record per customer interaction

Data Dictionary

Column Name	Data Type	Description	Sample / Allowed Values	Business Rules and Notes
Interaction ID	String	Unique identifier for each ad interaction or site visit.	INT0001, INT0002...	Primary key; must be unique across all rows.
Campaign Type	Categorical	The primary strategic messaging creative used in the ad.	Sales-Focused, Brand-Focused (A or B)	Used to compare performance between promotional and brand-building content.
Channel	Categorical	The marketing medium or platform where the interaction occurred.	Email, Instagram, Web banner	Used to evaluate acquisition efficiency across platforms.
Customer Type	Categorical	Classification of the customer based on historical engagement.	New, Existing	Critical for isolating new customer acquisition drivers vs. retention.

Converted	Binary (Flag)	Indicates whether the interaction resulted in a completed purchase action.	1 (Converted / Yes) 0 (Not Converted / No)	Numerical flag used to easily calculate Conversion rates
Time on Site (seconds)	Continuous	Duration of the user's session in seconds following the ad click.	45, 120, 310	Key engagement metric; non-converters may still log session time.
Sales (\$)	Continuous	Total revenue generated from the interaction (in USD).	0.00, 45.50, 129.99	Value is \$0.00 if Converted = 0. Total sales equal revenue from converted transactions.

Data Audit and Pivot Table Summary

Standard Pivot Tables

- **Total Sales by Channel:** Aggregates Sales (\$) grouped by Channel to evaluate total revenue contribution per marketing medium.
- **Total Sales by Campaign:** Aggregates Sales (\$) grouped by Campaign Type to determine revenue generated by sales-focused vs. brand-focused messaging.

Cross-Dimensional Pivot Table (Channel x Campaign)

To evaluate the combined impact of creative strategy and delivery medium, construct a primary 2D matrix:

- **Rows:** Channel
- **Columns:** Campaign Type
- **Values:** Sum of Sales (\$), Count of Interaction ID, and Average Conversion Rate (Average of Converted)

Segment Filtering (New Customer Slicer)

- **Slicer Applied:** Customer Type filtered strictly to **New**.
- **Analytical Purpose:** Isolates high-intent acquisition drivers, identifying which specific channel and campaign combinations attract and convert first-time buyers vs. returning users.